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# MedShield AI - 3-Laptop Network Setup Guide

## Quick Start

### Step 1 - Connect all 3 laptops to same WiFi
All 3 must be on the same network (hotspot or router).

### Step 2 - Find each laptop's IP
Open CMD on each laptop and run:
```
ipconfig
```
Look for "IPv4 Address" under your WiFi adapter.

### Step 3 - Update config.py on ALL 3 laptops
Open `config.py` and set the real IPs:
```python
PUMP_IP = "192.168.1.10" # Laptop 1 actual IP
ATTACKER_IP = "192.168.1.20" # Laptop 2 actual IP
MEDSHIELD_IP = "192.168.1.30" # Laptop 3 actual IP
```

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## Laptop Roles

### Laptop 1 - i5 13th Gen + RTX 4050
**Role: Insulin Pump Simulator (Victim)**
```
Files needed:
app.py ← Flask pump server
dashboard.py ← PyQt6 visual dashboard
config.py ← Network config
requirements.txt

How to run:
Double-click: run_laptop1.bat
OR manually:
Terminal 1: python app.py
Terminal 2: python dashboard.py
```

### Laptop 2 - i5 12th Gen + RTX 3050
**Role: Attacker Machine**
```
Files needed (build next):
attacks/
 attack_1_overdose.py
 attack_2_rapid_repeat.py
 attack_3_spoof.py
 attack_4_pairing.py
 attack_5_credential.py
 attack_launcher.py ← menu to pick attack
config.py

How to run:
python attacks/attack_launcher.py
Choose target: pump (no MedShield) or medshield (with MedShield)
```

### Laptop 3 - i7 13th Gen + RTX 4050
**Role: MedShield AI (Defender)**
```
Files needed (build next):
main.py ← FastAPI proxy server
pipeline/
 11_zero_trust.py
 12_rule_engine.py
 13_anomaly.py
 14_fingerprint.py
 15_behaviour.py
 16_digital_twin.py
 17_xai.py
 18_trust.py
esc.py
models/
 isolation_forest.pkl
 autoencoder.h5
 lstm_twin.h5
dashboard/ ← MedShield live monitor
config.py

```